

Git and GitHub

Submitted to Vikul_Sir

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L2 - Create Local git repository and demonstrate git merge and Merge Conflicts with the steps to resolve merge conflicts

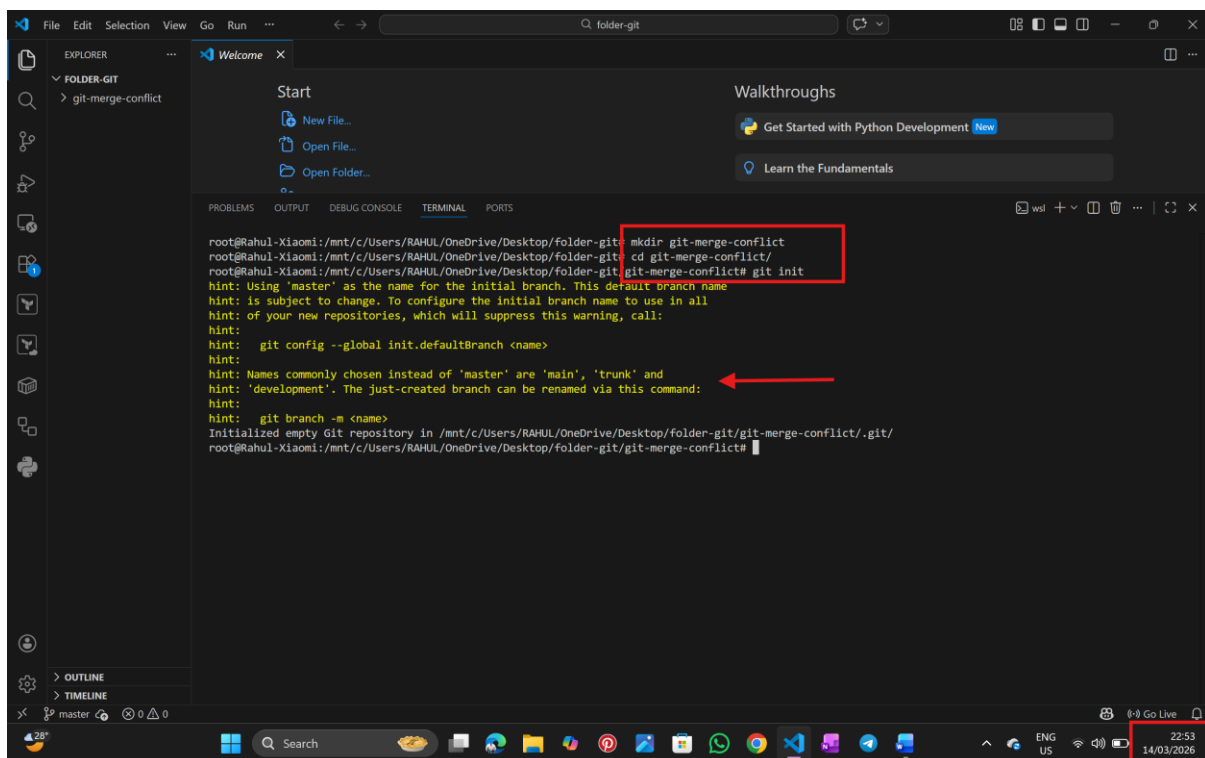
Step: 1 open visual studio code

→create a local git repository

→make a directory use command – mkdir git-merge-demo

→change to the directory use command – cd git-merge-demo

→initialization of git use command – git init

A screenshot of the Visual Studio Code interface. The Explorer panel on the left shows a folder named 'FOLDER-GIT' with a subfolder 'git-merge-conflict'. The main editor area displays the 'Welcome' page. The Terminal panel at the bottom shows a series of commands and their outputs. The commands executed are: 'mkdir git-merge-conflict', 'cd git-merge-conflict', and 'git init'. The output of 'git init' shows a warning about the 'master' branch name and suggests using 'main' or 'trunk' instead. A red box highlights the 'git init' command and its output. A red arrow points to the hint text: 'Names commonly chosen instead of 'master' are 'main', 'trunk' and 'development''. The status bar at the bottom right shows the time as 22:53 on 14/03/2026.

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git# mkdir git-merge-conflict
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git# cd git-merge-conflict/
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git init
hint: Using 'master' as the name for the initial branch. This default 'branch name'
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict/.git/
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict#
```

Step: 2 create a file in the directory

→create a file use command echo “content” >> <filename> -- it creates a file and write content in it.

→and commit it

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# echo "this is python" >> file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# cat file.txt
this is python
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git status
On branch master

No commits yet

Untracked files:
  (use "git add -f" to include in what will be committed)
    file.txt

nothing added to commit but untracked files present (use "git add" to track)
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git add .
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   file.txt

root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git commit -m "add python"
[master (root-commit) 7257d40] add python
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git branch
* master
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict#
```

Step: 3 create a new branch

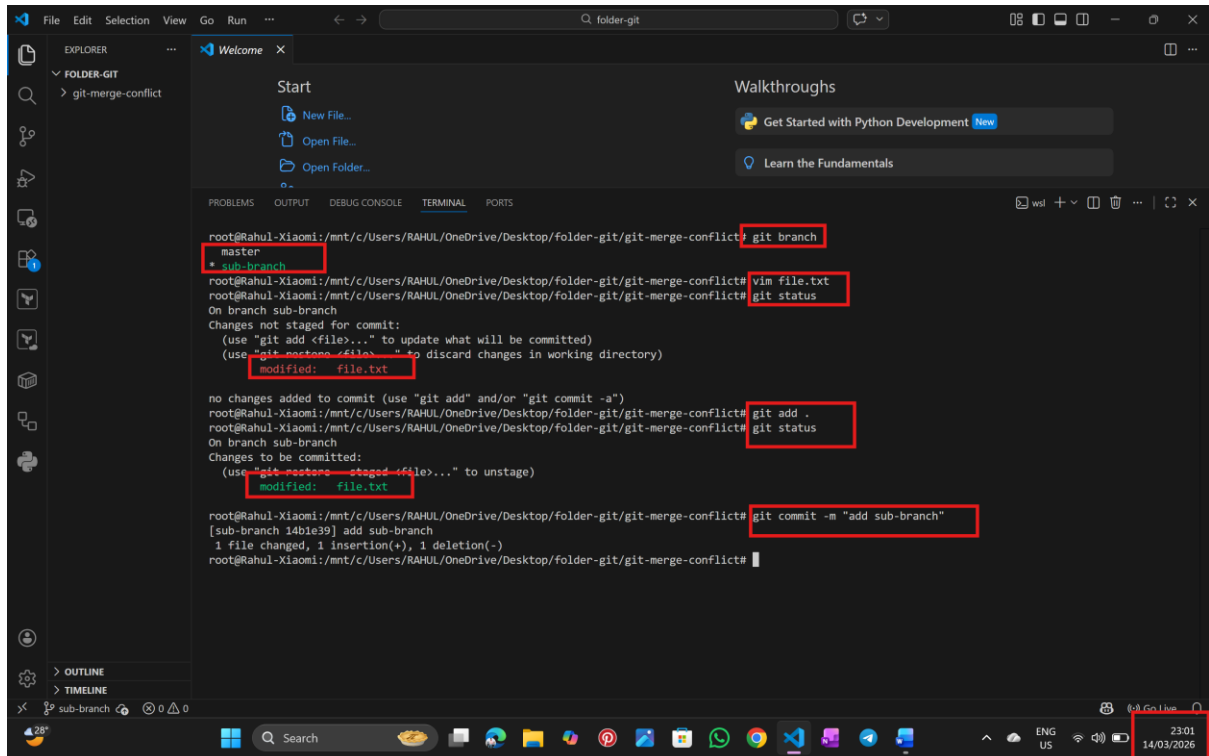
→ use command git switch -b <branch name> -- it will create branch and go inside the branch

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git branch
* master
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git checkout -b sub-branch
Switched to a new branch 'sub-branch'
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git branch
* sub-branch
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict#
```

Step: 4 now edit (or) change the content in the file you previously create

→use vim or nano or echo commands

→and commit



The screenshot shows a VS Code terminal window with the following commands and output:

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git branch
* master
* sub-branch
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# vim file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git status
On branch sub-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file.txt

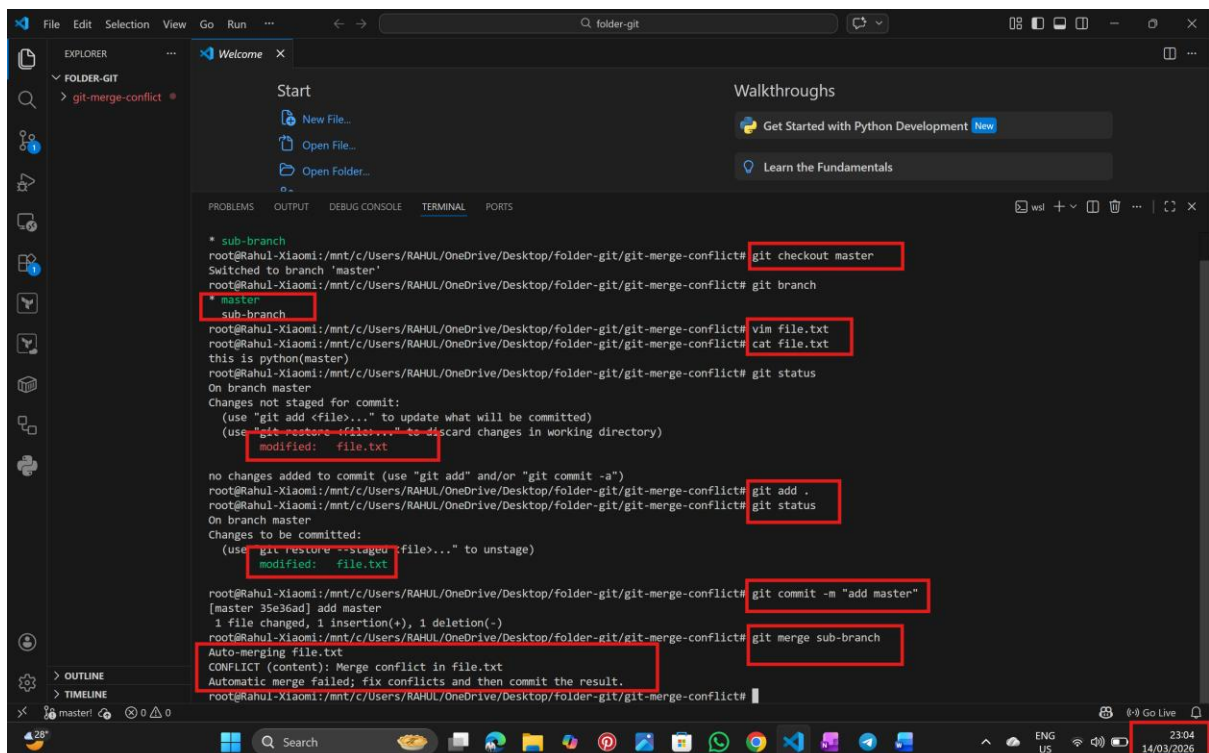
no changes added to commit (use "git add" and/or "git commit -a")
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git add .
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git status
On branch sub-branch
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   file.txt

root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git commit -m "add sub-branch"
[sub-branch 14b1e29] add sub-branch
1 file changed, 1 insertion(+), 1 deletion(-)
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict#
```

The terminal window is titled "Welcome" and shows the "git-merge-conflict" folder in the Explorer. The status bar at the bottom indicates the current branch is "sub-branch" and the time is 23:01 on 14/03/2026.

Step: 5 now switch back to main branch and merge conflict

→and edit the content in the same →commit it →and use git merge <branchname>



The screenshot shows a VS Code terminal window with the following commands and output:

```
* sub-branch
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git checkout master
Switched to branch 'master'
* master
* sub-branch
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# vim file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# cat file.txt
this is python(master)
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file.txt

no changes added to commit (use "git add" and/or "git commit -a")
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git add .
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   file.txt

root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git commit -m "add master"
[master 35e36ad] add master
1 file changed, 1 insertion(+), 1 deletion(-)
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git merge sub-branch
Auto-merging file.txt
CONFLICT (content): Merge conflict in file.txt
Automatic merge failed; fix conflicts and then commit the result.
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict#
```

The terminal window is titled "Welcome" and shows the "git-merge-conflict" folder in the Explorer. The status bar at the bottom indicates the current branch is "master" and the time is 23:04 on 14/03/2026.

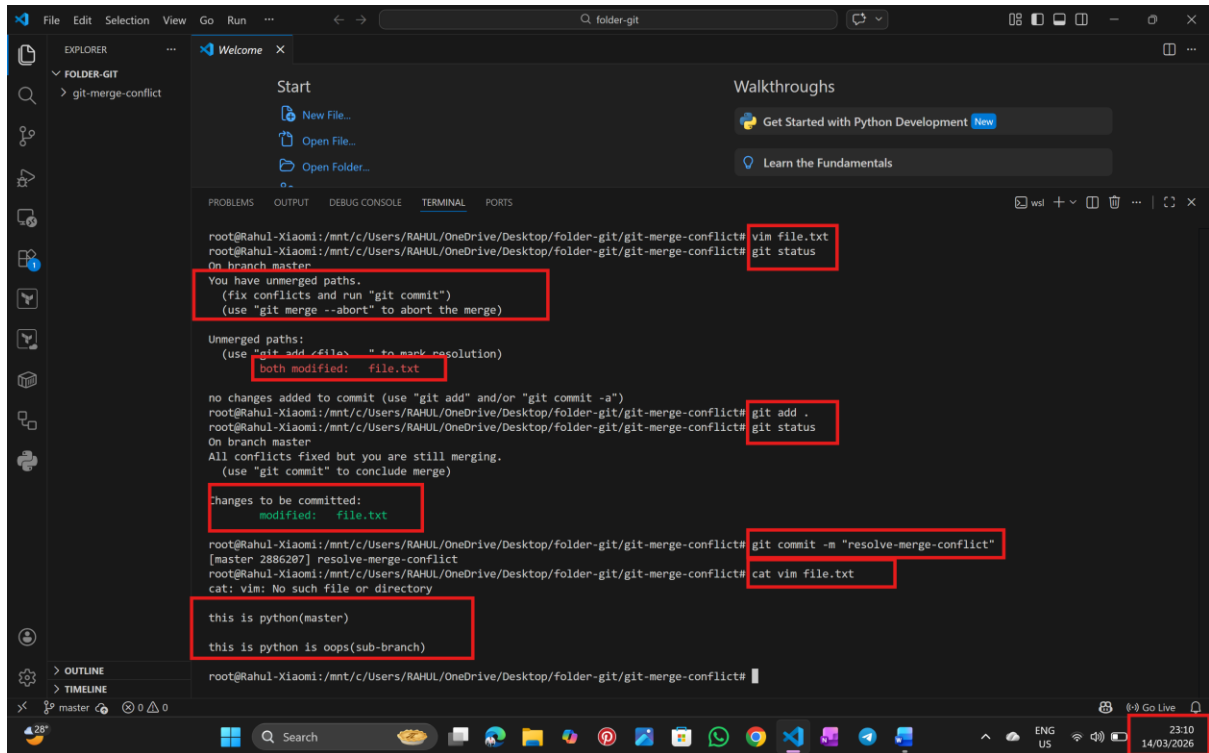
Step: 7 To resolve the merge conflict

→ it is resolved manually

→ open the file and delete the lines (HEAD LINE, MIDDLE LINE AND END LINE)

→ save the file use ESC -- :wq!

→ and commit it



```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# vim file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>" to mark resolution)
    both modified:   file.txt

no changes added to commit (use "git add" and/or "git commit -a")
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git add .
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git status
On branch master
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   file.txt

root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# git commit -m "resolve-merge-conflict"
[master 2886207] resolve-merge-conflict
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict# cat vim file.txt
cat: vim: No such file or directory

this is python(master)
this is python is oops(sub-branch)

root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-merge-conflict#
```

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NALATHATI RAHUL